

Workflow labs use the variant template: Goal → Approach → Execution → Check → Report.

Learning objectives

- Match the right summary statistic to the right variable type — means for symmetric continuous, medians for skewed, counts and percentages for categorical.
- Build a publication-ready Table 1 with `gtsummary::tbl_summary()`.
- Defend the choice of summary with a quick visual check.

Prerequisites

Labs 1.3 through 1.5.

Background

Table 1 is the first real table of almost every clinical paper. It describes the sample: who was in the study, in how many groups, with what characteristics, and — if the study has arms — how balanced those arms are at baseline. A Table 1 that is built thoughtlessly hides an unbalanced comparison; a Table 1 that is built well anchors the rest of the paper.

The right summary depends on the variable. A continuous variable that is roughly symmetric is summarised by its mean and standard deviation; one that is skewed or has heavy tails is summarised by its median and interquartile range. A categorical variable is summarised by counts and percentages. Reporting the median of a balanced continuous variable is not wrong, but it is unusual; reporting the mean of a skewed variable is often misleading.

`gtsummary` packages these choices into a declarative API. You say which variables to include and which grouping variable to stratify by, and the table is built with sensible defaults. When you need non-standard behaviour, you override per-variable.

Setup

```
library(tidyverse)
library(palmerpenguins)
library(gtsummary)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Produce a Table 1 for the `penguins` dataset stratified by species, and back the choice of summary with a diagnostic plot.

2. Approach

```
drop_na(species, sex, bill_length_mm, bill_depth_mm,
         flipper_length_mm, body_mass_g)
```

3. Execution

Pick the right summary per variable. First, eyeball the shapes.

```
dat |>
  pivot_longer(c(bill_length_mm, bill_depth_mm,
                 flipper_length_mm, body_mass_g),
```

```

      names_to = "variable", values_to = "value") |>
ggplot(aes(value, fill = species)) +
geom_histogram(bins = 20, alpha = 0.6, colour = "grey30",
               position = "identity") +
facet_wrap(~ variable, scales = "free") +
labs(x = NULL, y = "Count")

```

All four continuous variables are approximately symmetric within species, so mean (SD) is defensible. If a variable were skewed, we would switch to median (IQR).

```

select(species, sex, bill_length_mm, bill_depth_mm,
       flipper_length_mm, body_mass_g) |>
tbl_summary(
  by = species,
  statistic = list(
    all_continuous() ~ "{mean} ({sd})",
    all_categorical() ~ "{n} ({p}%)"
  ),
  digits = all_continuous() ~ 1,
  label = list(
    bill_length_mm ~ "Bill length (mm)",
    bill_depth_mm ~ "Bill depth (mm)",
    flipper_length_mm ~ "Flipper length (mm)",
    body_mass_g ~ "Body mass (g)",
    sex ~ "Sex"
  )
) |>
add_overall() |>
add_p()
t1

```

`add_p()` will run a t-test or ANOVA on continuous variables and a chi-square or Fisher's exact test on categorical, with sensible defaults for sample size.

4. Check

Validate the `gtsummary` output against a hand computation.

```

group_by(species) |>
summarise(
  n = n(),
  mean_mass = mean(body_mass_g),
  sd_mass = sd(body_mass_g),
  .groups = "drop"
)

```

The group means and standard deviations must match the numbers in the body-mass row of the Table 1.

5. Report

Table 1 describes `nrow(dat)` penguins from three species in the Palmer Archipelago. Body mass was highest in Gentoo ($\text{round}(\text{mean}(\text{dat}\$body_mass_g[\text{dat}\$species == "Gentoo"])$, 0) $\pm$ $\text{round}(\text{sd}(\text{dat}\$body_mass_g[\text{dat}\$species == "Gentoo"])$, 0) g) and lower in Adelie and Chinstrap. All four continuous variables were approximately symmetric within species and are summarised as mean (SD); sex was approximately balanced within each species.

Table 1 is not a tool for hypothesis testing. The p -values in the rightmost column describe the data, not the biology. A statistically-significant difference between arms at baseline in a randomised trial is, by construction, a chance finding.

Common pitfalls

- Reporting a mean for a skewed variable.
- Using Table 1 as a place to “prove balance” with p -values in a randomised trial.
- Suppressing the missing-data row. Always report it.
- Forgetting that `tbl_summary` drops NAs by default; check the count at the bottom of each column.

Further reading

- Sjöberg DD et al. *Reproducible summary tables with the gtsummary package*. R Journal.
- Altman DG. *Practical Statistics for Medical Research*, chapter on describing data.

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)